

# Timeline Check-In

## Set Up

### Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# read .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)

library(dplyr)      # data manipulation
library(broom)      # tidy statistical output
library(corrplot)
```

### Data

```

# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# read in site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# read in water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

```

## Cleaning

### sites object

```

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

### Cleaning taxon\_list

```

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

```

## Cleaning water\_quality

```
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(
    med_ph = median(dissolved_oxygen_mg_l, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

## Cleaning aq\_ins

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(
    date, sample_type, site,
    ostracod,
    copepod,
    hemiptera_corixidae_boatman,
    cladocera,
    nematode,
    diptera_ceratopogonidae,
    annelida_oligochaete,
```

```

    diptera_chironomid,
    annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5 liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_do, .fun = "median", .na_rm = TRUE),
       site_full = fct_rev(site_full))

```

### Focus Taxa (focus\_taxa)

```

# taxa to include in all figures
focus_taxa <- c("Ostracod", "Copepod", "Cladocera",
                "Hemiptera Corixidae Boatman", "Nematode")

```

## Creating DO bins (creating do\_category column)

```
aq_ins_clean_bins <- aq_ins_clean |>
  mutate(bins = case_when(site_full == "Phelps Road" ~ "low",
                           site_full == "North of Phelps Bridge" ~ "low",
                           site_full == "South of Venoco Bridge" ~ "moderate",
                           site_full == "South of Phelps Bridge" ~ "moderate",
                           site_full == "Devereux Creek" ~ "moderate",
                           site_full == "North of Venoco Bridge" ~ "high",
                           site_full == "Main Channel" ~ "high")) |>
  mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
  filter(taxon %in% focus_taxa) |>
  # create column that calculates log-transformed abundance
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  mutate(bins = factor(bins, levels = c("low", "moderate", "high")))

# create object DO_taxa from aq_ins_clean
DO_taxa <- aq_ins_clean |>
  # remove rows that are missing median salinity values
  filter(!is.na(med_do)) |>
  # create column that calculates log-transformed abundance
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # mutates to title-case taxon names
  mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
  # only include focus taxa
  filter(taxon %in% focus_taxa) |>
  # order taxon levels by maximum average abundance per liter
  mutate(taxon = fct_reorder(taxon, ave_lit, .fun = max, .desc = TRUE)) |>
  # keep only the columns needed for the figure
  select(date, site, site_full, taxon, ave_lit, log_ave_lit, med_do)
total_abundance <- DO_taxa |>
  filter(taxon %in% focus_taxa) |>
  group_by(date, site, med_do) |>
  summarize(
    total_abundance = sum(ave_lit, na.rm = TRUE),
    log_total_abundance = log10(total_abundance + 1),
    .groups = "drop"
  )
total_abundance <- total_abundance |>
  mutate(
    do_category = case_when(
      med_do <= quantile(med_do, 0.33, na.rm = TRUE) ~ "Low DO",
```

```

    med_do <= quantile(med_do, 0.66, na.rm = TRUE) ~ "Medium DO",
    TRUE ~ "High DO"
  )
)
total_abundance$do_category <- factor(
  total_abundance$do_category,
  levels = c("Low DO", "Medium DO", "High DO")
)

```

## Ideas for analysis and background

- should include some citations (not all)
- should include ideas for analyses to answer questions and some preliminary code to run those analyses

### Citations:

NEED TO ADD

## Spearman Correlation Analysis for Total Abundance and DO levels

```

cor.test(
  total_abundance$log_total_abundance,
  total_abundance$med_do,
  method = "spearman",
  exact = FALSE)

```

Spearman's rank correlation rho

data: total\_abundance\$log\_total\_abundance and total\_abundance\$med\_do

S = 8526.3, p-value = 0.08182

alternative hypothesis: true rho is not equal to 0

sample estimates:

rho  
-0.3027201

## Correlation analysis for Each Focus Taxa and DO levels

```
# create wide dataset (one column per taxon)
taxa_wide <- DO_taxa |>
  select(date, site, taxon, log_ave_lit, med_do) |>
  pivot_wider(
    names_from = taxon,
    values_from = log_ave_lit
  )

# calculate correlation of each taxon with dissolved oxygen
# calculate Spearman correlation of each taxon with dissolved oxygen
cor_do <- taxa_wide |>
  select(-date, -site) |>
  summarize(across(-med_do, ~ cor(.x,
                                med_do,
                                method = "spearman",
                                use = "complete.obs"))) |>
  pivot_longer(cols = everything(),
    names_to = "taxon",
    values_to = "correlation")

ggplot(cor_do, aes(x = taxon, y = "Dissolved oxygen", fill = correlation)) +
  geom_tile(color = "grey70") +
  geom_point(aes(size = abs(correlation)), shape = 21, fill = "white") +

  # move labels ABOVE circles
  geom_text(aes(label = round(correlation, 2)),
    nudge_y = 0.15, # <-- this moves text up
    size = 4) +

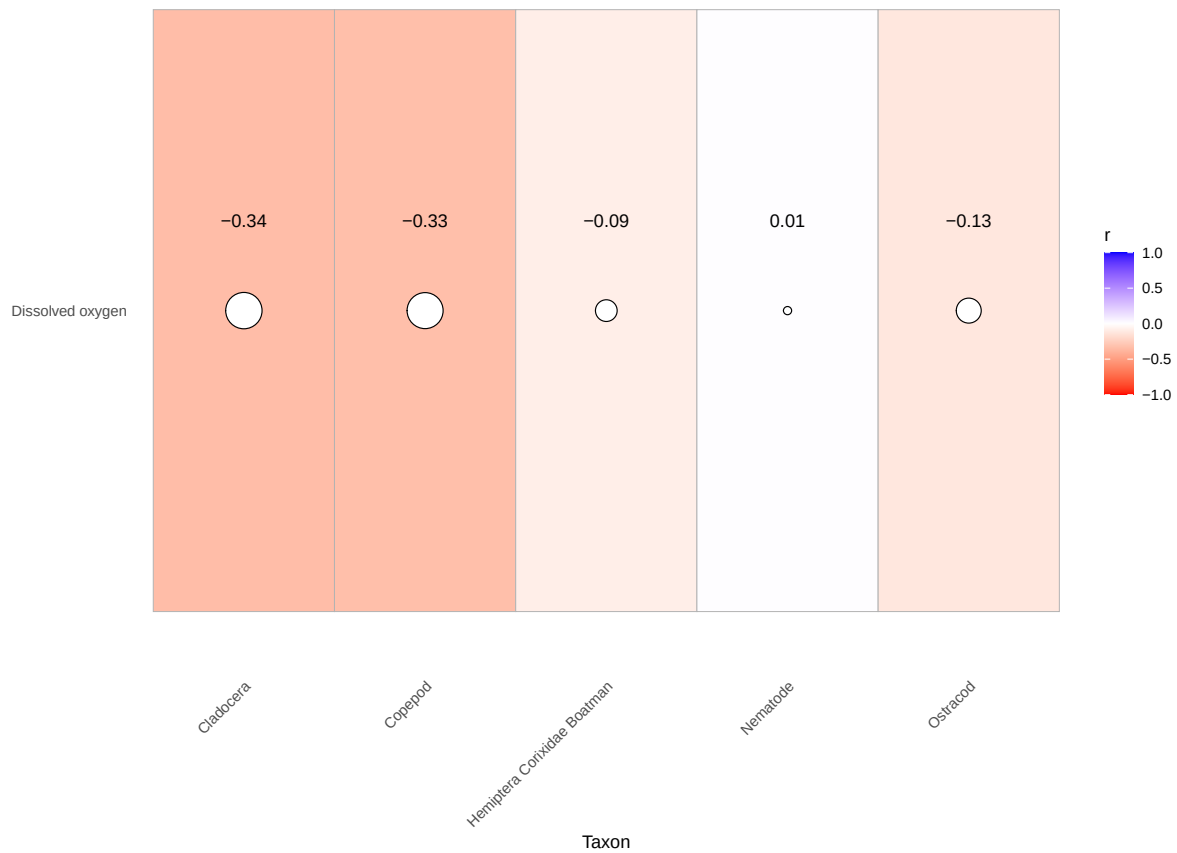
  scale_fill_gradient2(
    low = "red",
    mid = "white",
    high = "blue",
    midpoint = 0,
    limits = c(-1, 1),
    name = "r"
  ) +
  scale_size(range = c(2, 10), guide = "none") +
  labs(
    x = "Taxon",
```

```

y = NULL,
title = "Spearman correlation between dissolved oxygen and invertebrate abundance"
) +
theme_minimal() +
theme(
  panel.grid = element_blank(),
  axis.text.x = element_text(angle = 45, hjust = 1)
)

```

Spearman correlation between dissolved oxygen and invertebrate abundance



### Pearson correlation for continuous DO and total community abundance

```

# Pearson correlation for continuous DO and total community abundance
cor.test(total_abundance$log_total_abundance,

```



```
total_abundance$med_do,
method = "pearson")
```

Pearson's product-moment correlation

```
data: total_abundance$log_total_abundance and total_abundance$med_do
t = -2.2882, df = 32, p-value = 0.02887
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.63289820 -0.04217141
sample estimates:
      cor
-0.3749894
```

```
# r = -0.3749894, p = 0.02887
# there is a significant negative correlation between abundance and continous DO
```

### Kruskal-Wallis test comparing abundance across DO categories

```
kruskal.test(log_total_abundance ~ do_category,
             data = total_abundance)
```

Kruskal-Wallis rank sum test

```
data: log_total_abundance by do_category
Kruskal-Wallis chi-squared = 3.2212, df = 2, p-value = 0.1998
```

```
# Kruskal-Wallis chi-squared = 3.2212, df = 2, p-value = 0.1998
# there is not a significant difference between abundance and DO categories
```

### Kruskal-Wallis test between abundance and bins for Copepods

```
kruskal.test(log_ave_lit ~ bins,
             data = aq_ins_clean_bins,
             subset = taxon == "Copepod") # Kruskal-Wallis chi-squared = 4.405, df = 2, p-value = 0.108
```

Kruskal-Wallis rank sum test

data: log\_ave\_lit by bins

Kruskal-Wallis chi-squared = 4.405, df = 2, p-value = 0.1105

# no significant difference

## Kruskal-Wallis test between abundance and bins for Ostracods

```
kruskal.test(log_ave_lit ~ bins,
             data = aq_ins_clean_bins,
             subset = taxon == "Ostracod") # Kruskal-Wallis chi-squared = 6.7235, df = 2, p-value = 0.03467
```

Kruskal-Wallis rank sum test

data: log\_ave\_lit by bins

Kruskal-Wallis chi-squared = 6.7235, df = 2, p-value = 0.03467

# significant difference of abundance btwn DO levels

## Pairwise test between abundance and individual bins for Ostracods

```
bins_ostracods <- aq_ins_clean_bins |>
  filter(taxon == "Ostracod")
pairwise.wilcox.test(
  bins_ostracods$log_ave_lit,
  bins_ostracods$bins,
  p.adjust.method = "bonferroni") # p = 0.051 for comparison btwn moderate and high DO level
```

Pairwise comparisons using Wilcoxon rank sum test with continuity correction

data: bins\_ostracods\$log\_ave\_lit and bins\_ostracods\$bins

low      moderate

```
moderate 0.387 -
high      1.000 0.051
```

P value adjustment method: bonferroni

## Chi-Square

```
# create abundance table for chi-square test
chi_square_table <- aq_ins_clean_bins |>
  filter(taxon %in% focus_taxa) |>
  group_by(bins, taxon) |>
  summarise(
    total_abundance = sum(ave_lit, na.rm = TRUE),
    .groups = "drop"
  ) |>
  pivot_wider(
    names_from = taxon,
    values_from = total_abundance,
    values_fill = 0
  ) |>
  column_to_rownames("bins") |>
  as.matrix()

# show table used in chi-square test
chi_square_table
```

|          | Cladocera | Copepod   | Hemiptera | Corixidae | Boatman    | Nematode  | Ostracod    |
|----------|-----------|-----------|-----------|-----------|------------|-----------|-------------|
| low      | 980.2667  | 115.8667  |           |           | 0.2666667  | 0.1333333 | 1202.933333 |
| moderate | 510.6667  | 1170.4000 |           |           | 83.8666667 | 0.2666667 | 1057.866667 |
| high     | 264.2667  | 1453.4667 |           |           | 8.8000000  | 0.0000000 | 7.866667    |

```
# chi-square test for differences in taxon composition across DO bins
chi_square_test <- chisq.test(chi_square_table)

# show chi-square result
chi_square_test
```

Pearson's Chi-squared test

```
data: chi_square_table
X-squared = 2859.2, df = 8, p-value < 2.2e-16
```

## Visualizations directly relevant to answering questions

### Directly Relevant Figure 1: DO levels vs invert. abundance for each focus taxa

```
DO_taxa_medians <- DO_taxa |>
  # groups data by taxon so each species gets one median point
  group_by(taxon) |>
  # calculates median DO and median log abundance for each taxon
  summarise(
    med_do_point = median(med_do, na.rm = TRUE),
    med_log_ave_lit = median(log_ave_lit, na.rm = TRUE),
    .groups = "drop"
  )

DO_taxa_plot <- ggplot(data = DO_taxa, # assigns plot to object do_taxa_plot
  # uses DO_taxa data frame
  mapping = aes(x = med_do, # x-axis: med_do
    y = log_ave_lit, # y-axis: log_ave_lit
    color = taxon)) + # color by taxon

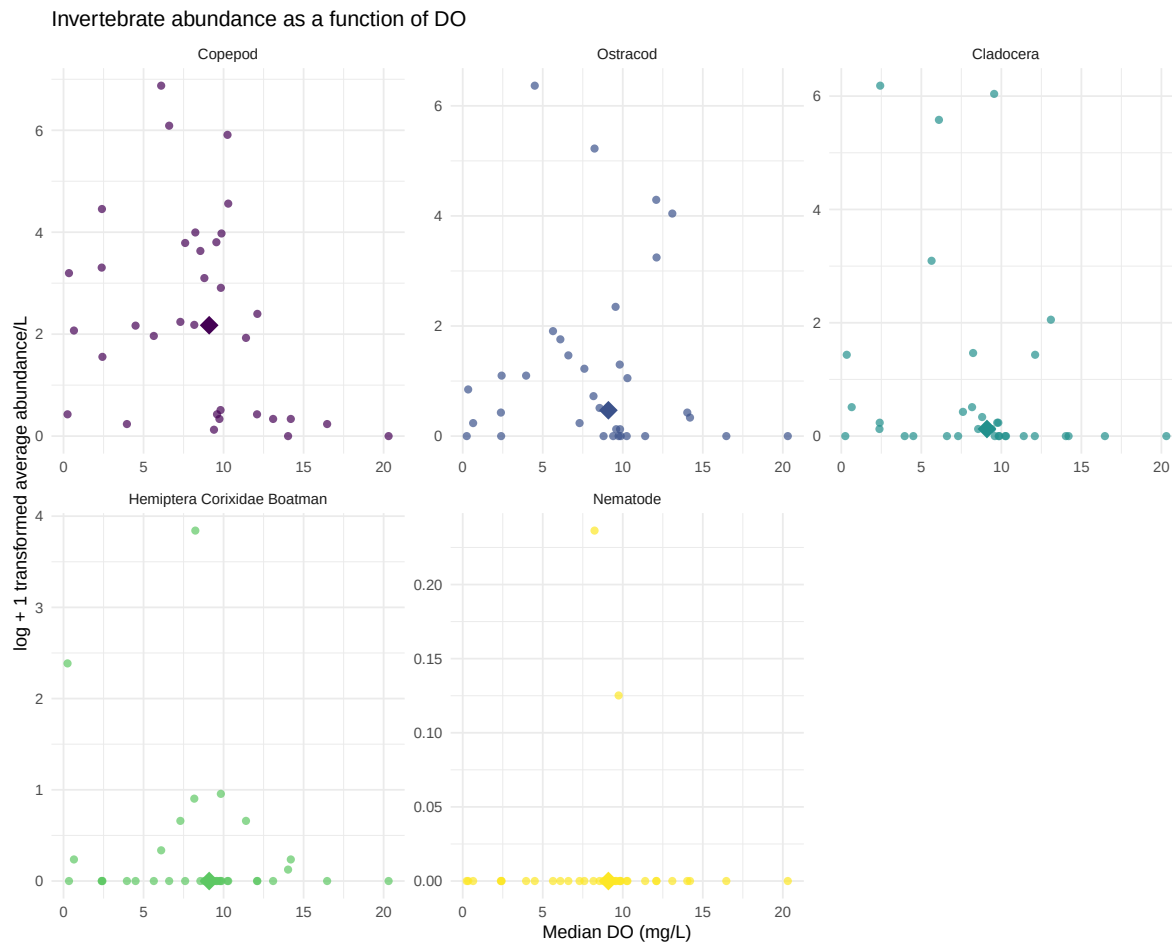
  # first layer: points show individual abundance measurements
  geom_point(shape = 16, # shape is circle
    size = 2, # size is 2
    alpha = 0.7) + # opacity is 0.7

  # second layer: larger point shows the median location for each taxon
  geom_point(data = DO_taxa_medians,
    mapping = aes(x = med_do_point,
      y = med_log_ave_lit),
    shape = 18,
    size = 5) +

  # separate panels for each taxon with free x- and y-axis scales
  facet_wrap(~ taxon, scales = "free") +
  # use non-default colors for points
  scale_color_viridis_d() +
  labs(x = "Median DO (mg/L)", # relabel x-axis
    y = "log + 1 transformed average abundance/L", # relabel y-axis
```

```
# creates descriptive title
title = "Invertebrate abundance as a function of DO") +

# uses non-default theme
theme_minimal() +
# remove redundant legend
theme(legend.position = "none")
DO_taxa_plot
```



**Directly Relevant Figure 2: DO levels vs total invert. abundance**

```
# Summed abundance across focus taxa vs DO
focus_taxa <- c("Cladocera", "Copepod",
```

```

      "Hemiptera Corixidae Boatman",
      "Nematode", "Ostracod")

total_abundance <- DO_taxa |>
  filter(taxon %in% focus_taxa) |>
  group_by(date, site, med_do) |>
  summarize(
    total_abundance = sum(ave_lit, na.rm = TRUE),
    .groups = "drop"
  ) |>
  mutate(log_total_abundance = log10(total_abundance + 1))

# fit linear model
total_abundance_lm <- lm(log_total_abundance ~ med_do,
                        data = total_abundance)

# create new data frame for predicted values
total_abundance_predictions <- data.frame(
  med_do = seq(min(total_abundance$med_do, na.rm = TRUE),
               max(total_abundance$med_do, na.rm = TRUE),
               length.out = 100)
)

# generate predicted values and confidence intervals
total_abundance_predictions <- total_abundance_predictions |>
  bind_cols(
    predict(total_abundance_lm,
            newdata = total_abundance_predictions,
            interval = "confidence") |>
    as.data.frame()
  )

# plot raw data and model predictions
ggplot(total_abundance,
       aes(x = med_do,
           y = log_total_abundance)) +

# raw data points
geom_point(color = "magenta3",
           size = 3,
           alpha = 0.7) +

```

```

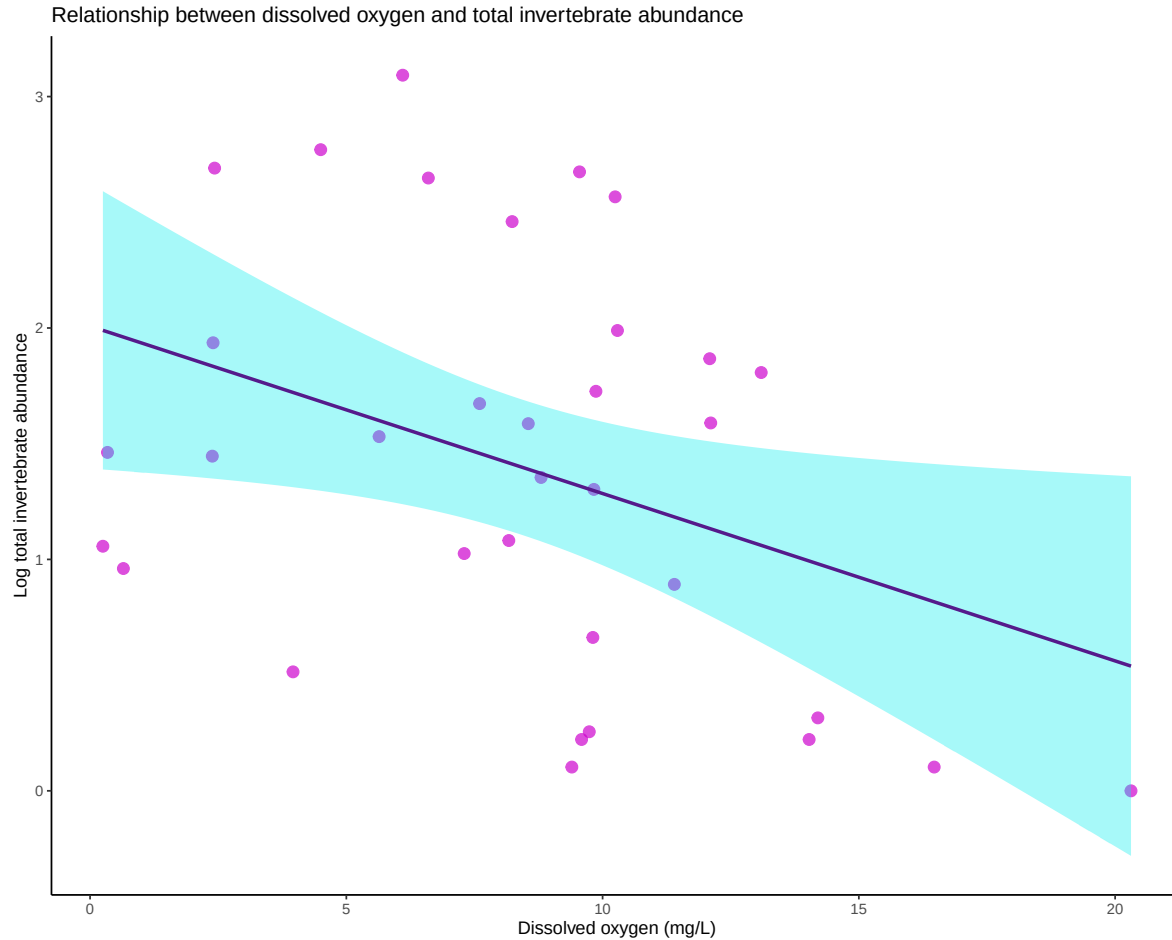
# confidence interval around model predictions
geom_ribbon(data = total_abundance_predictions,
           aes(x = med_do,
               ymin = lwr,
               ymax = upr),
           inherit.aes = FALSE,
           fill = "cyan2",
           alpha = 0.35) +

# predicted line from linear model
geom_line(data = total_abundance_predictions,
          aes(x = med_do,
              y = fit),
          inherit.aes = FALSE,
          color = "purple4",
          linewidth = 1) +

labs(
  x = "Dissolved oxygen (mg/L)",
  y = "Log total invertebrate abundance",
  title = "Relationship between dissolved oxygen and total invertebrate abundance"
) +

theme_classic()

```



```
broom::tidy(total_abundance_lm)
```

```
# A tibble: 2 x 5
```

|   | term        | estimate | std.error | statistic | p.value     |
|---|-------------|----------|-----------|-----------|-------------|
|   | <chr>       | <dbl>    | <dbl>     | <dbl>     | <dbl>       |
| 1 | (Intercept) | 2.01     | 0.302     | 6.65      | 0.000000170 |
| 2 | med_do      | -0.0723  | 0.0316    | -2.29     | 0.0289      |

### Directly Relevant Figure 3: Abundance vs DO with binned levels

```
total_abundance <- total_abundance |>
  mutate(
    do_category = case_when(
```



```

    med_do <= quantile(med_do, 0.33, na.rm = TRUE) ~ "Low DO",
    med_do <= quantile(med_do, 0.66, na.rm = TRUE) ~ "Medium DO",
    TRUE ~ "High DO"
  )
)
total_abundance$do_category <- factor(
  total_abundance$do_category,
  levels = c("Low DO", "Medium DO", "High DO")
)

ggplot(total_abundance,
  aes(x = do_category,
      y = log_total_abundance,
      fill = do_category)) +

  geom_boxplot(alpha = 0.7,
    outlier.color = "black",
    outlier.size = 2) +

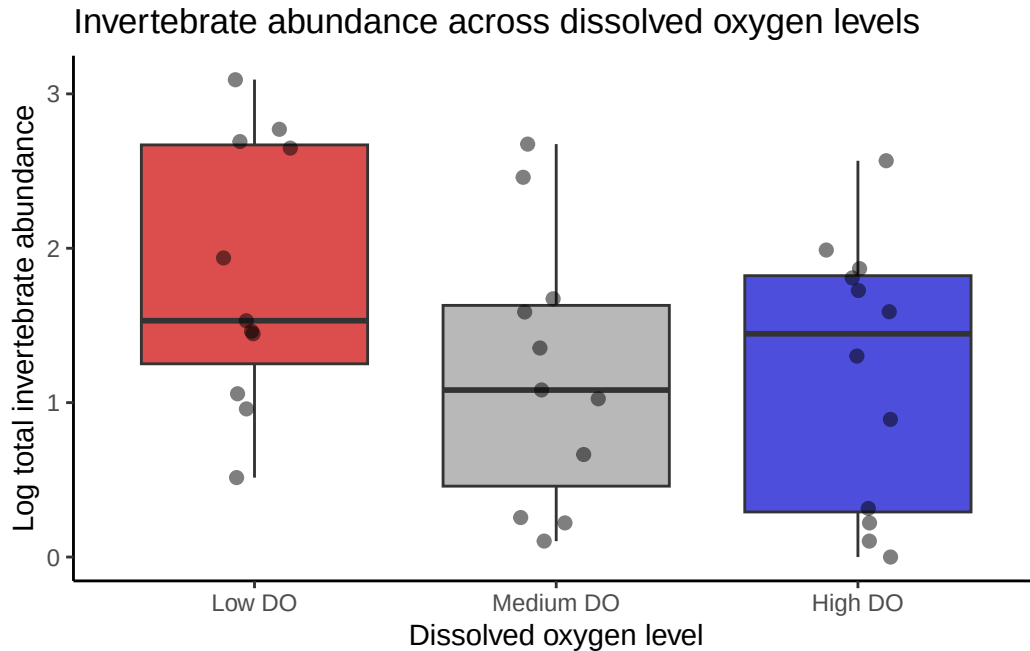
  # optional: add jittered points to show raw data
  geom_jitter(width = 0.15,
    alpha = 0.5,
    size = 2,
    color = "black") +

  labs(
    x = "Dissolved oxygen level",
    y = "Log total invertebrate abundance",
    title = "Invertebrate abundance across dissolved oxygen levels"
  ) +

  scale_fill_manual(values = c(
    "Low DO" = "red3",
    "Medium DO" = "grey60",
    "High DO" = "blue3"
  )) +

  theme_classic() +
  theme(legend.position = "none")

```



## Other exploratory visualizations

### Exploratory Figure 1: Variance of DO across sights

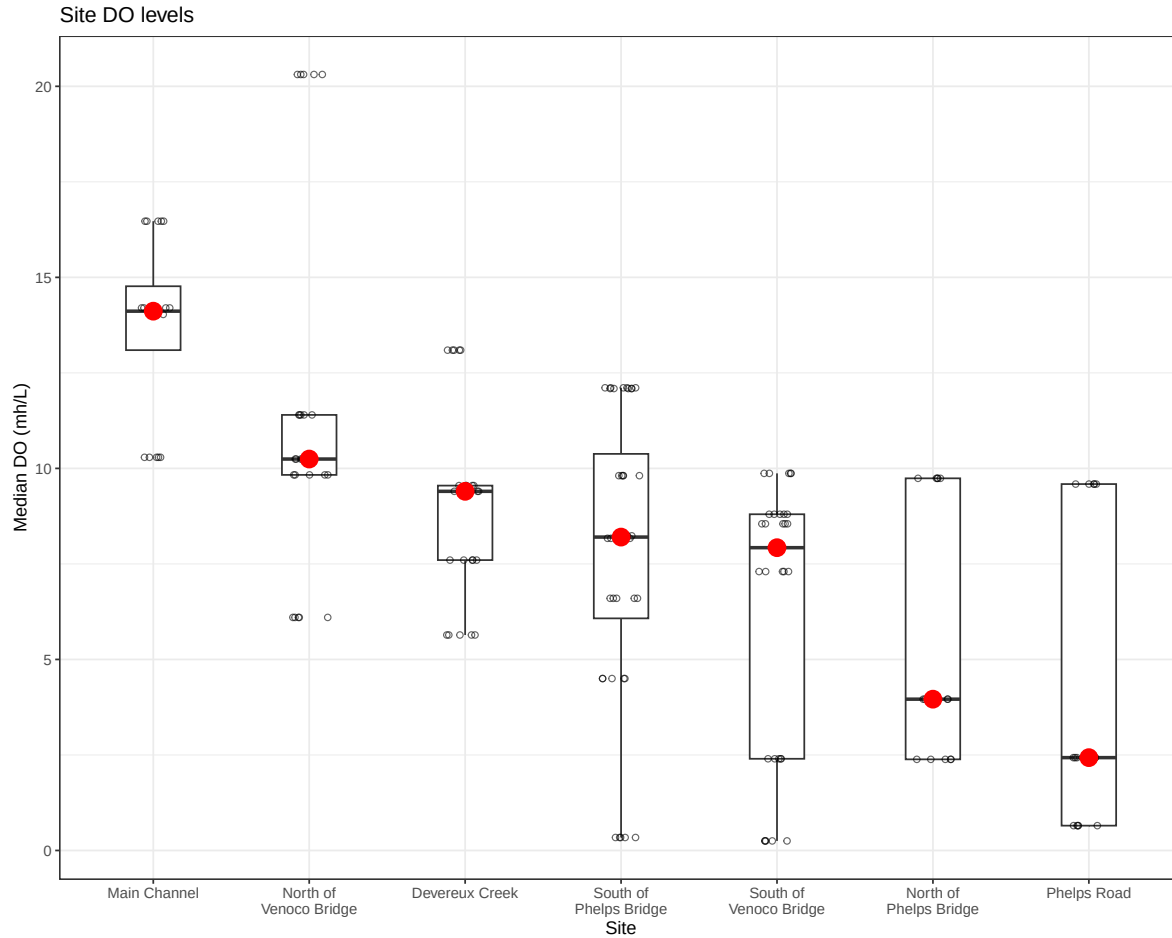
```
# create object aq_ins_clean_fig2 from aq_ins_clean
aq_ins_clean_fig2 <- aq_ins_clean |>
  # convert taxon to title case
  mutate(taxon = str_replace_all(taxon, "_", " "),
         taxon = str_to_title(taxon)) |>
  # filter to include focus taxa
  filter(taxon %in% focus_taxa)

# base layer
DO <- ggplot(data = aq_ins_clean_fig2,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
                y = med_do)) +
  # first layer: boxplot
  geom_boxplot(width = 0.35,
```

```

      outlier.shape = NA) +
# second layer: salinity measurements at each survey
geom_jitter(height = 0,
            width = 0.12,
            shape = 21,
            alpha = 0.6) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,
            color = "red",
            size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
     y = "Median DO (mh/L)",
     title = "Site DO levels") +
# different theme
theme_bw()
# show plot
DO

```



**Exploratory Figure 2: Composition plot across DO bins**

```
composition_plot <- DO_taxa |>
  mutate(
    do_category = case_when(
      med_do <= quantile(med_do, 0.33, na.rm = TRUE) ~ "Low DO",
      med_do <= quantile(med_do, 0.66, na.rm = TRUE) ~ "Medium DO",
      TRUE ~ "High DO"
    ),

    # reorder factor levels
    do_category = factor(do_category,
      levels = c("Low DO",
```

```

                                "Medium DO",
                                "High DO"))
) |>
group_by(do_category, taxon) |>
summarize(
  total_abundance = sum(ave_lit, na.rm = TRUE),
  .groups = "drop"
) |>
ggplot(aes(x = do_category,
           y = total_abundance,
           fill = taxon)) +

geom_col(position = "fill") +

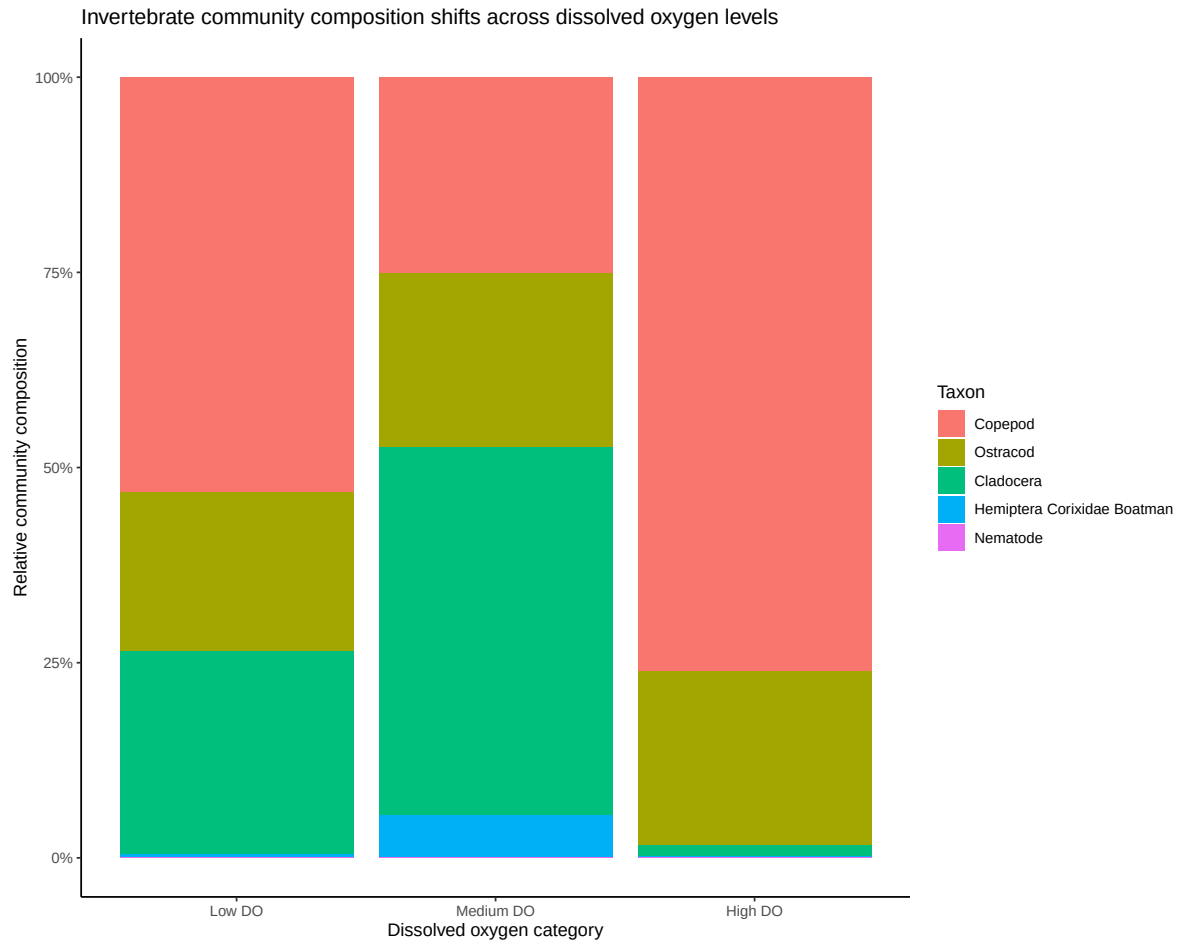
scale_y_continuous(labels = scales::percent_format()) +

labs(
  x = "Dissolved oxygen category",
  y = "Relative community composition",
  fill = "Taxon",
  title = "Invertebrate community composition shifts across dissolved oxygen levels"
) +

theme_classic()

composition_plot

```



**Exploratory Figure 3: Time-series plot of total abundance (all species)**

```
# species/taxa to include in the total abundance figure
species_interest <- c(
  "Ostracod",
  "Copepod",
  "Cladocera",
  "Hemiptera Corixidae Boatman",
  "Nematode"
)

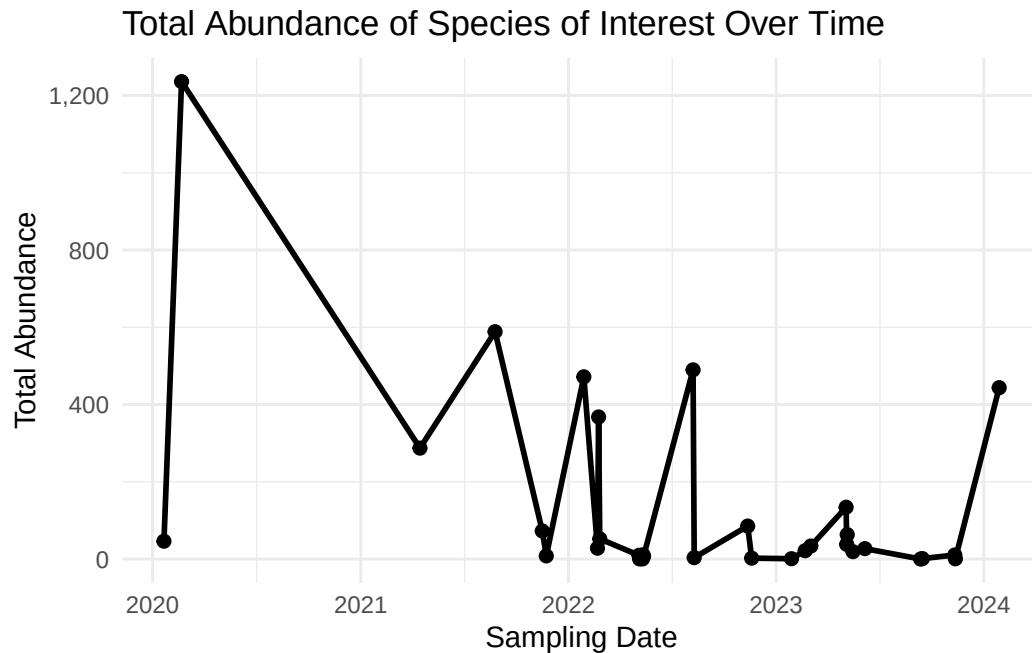
# create total abundance over time
total_abundance_time <- DO_taxa |>
```

```

# keeps only the taxa of interest
filter(taxon %in% species_interest) |>
# makes sure date is treated as a date
mutate(date = as.Date(date)) |>
# groups by date only, so abundance is summed across sites and taxa
group_by(date) |>
# sums average abundance per liter across all taxa of interest
summarise(
  total_abundance = sum(ave_lit, na.rm = TRUE),
  .groups = "drop"
)

# plot total abundance over time
ggplot(total_abundance_time, aes(x = date, y = total_abundance)) +
  # adds a line connecting total abundance values over time
  geom_line(linewidth = 1) +
  # adds points for each sampling date
  geom_point(size = 2) +
  # formats y-axis labels with commas
  scale_y_continuous(labels = scales::comma) +
  # adds clear labels
  labs(
    title = "Total Abundance of Species of Interest Over Time",
    x = "Sampling Date",
    y = "Total Abundance"
  ) +
  # uses a clean theme
  theme_minimal()

```



**Exploratory Figure 4: Time-series area plot**

```
# species/taxa to include in the area plot
species_interest <- c(
  "Ostracod",
  "Copepod",
  "Cladocera",
  "Hemiptera Corixidae Boatman",
  "Nematode"
)

# create abundance over time by taxon
taxon_abundance_time <- DO_taxa |>
  # keeps only the taxa of interest
  filter(taxon %in% species_interest) |>
  # makes sure date is treated as a date
  mutate(date = as.Date(date)) |>
  # groups by date and taxon so each species stays separate
  group_by(date, taxon) |>
  # sums across sites, but not across taxa
  summarise(
```

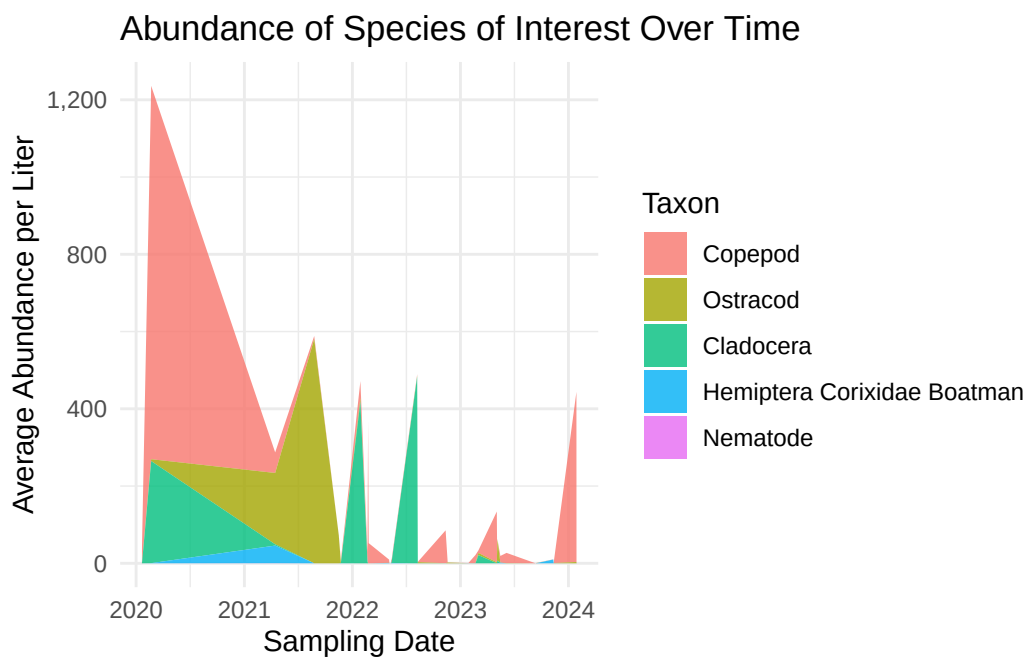


```

    ave_lit = sum(ave_lit, na.rm = TRUE),
    .groups = "drop"
  )

# create time series area plot by taxon
ggplot(taxon_abundance_time, aes(x = date, y = ave_lit, fill = taxon)) +
  # creates stacked area plot with a different color for each taxon
  geom_area(alpha = 0.8) +
  # formats y-axis labels with commas
  scale_y_continuous(labels = scales::comma) +
  # adds clear labels
  labs(
    title = "Abundance of Species of Interest Over Time",
    x = "Sampling Date",
    y = "Average Abundance per Liter",
    fill = "Taxon"
  ) +
  # uses a clean theme
  theme_minimal()

```



## Exploratory Figure 5: NEED TO ADD UPDATED FIGRUE HERE: Distribution of Median Dissolved Oxygen Aross Sites

## Exploratory Figure 6: Invertebrate Abundance in DO Bins

```
aq_ins_clean_bins <- aq_ins_clean |>
  mutate(bins = case_when(site_full == "Phelps Road" ~ "low",
                           site_full == "North of Phelps Bridge" ~ "low",
                           site_full == "South of Venoco Bridge" ~ "moderate",
                           site_full == "South of Phelps Bridge" ~ "moderate",
                           site_full == "Devereux Creek" ~ "moderate",
                           site_full == "North of Venoco Bridge" ~ "high",
                           site_full == "Main Channel" ~ "high")) |>
  mutate(taxon = str_replace_all(taxon, "_", " "), taxon = str_to_title(taxon)) |>
  filter(taxon %in% focus_taxa) |>
  # create column that calculates log-transformed abundance
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  mutate(bins = factor(bins, levels = c("low", "moderate", "high")))

# base layer: ggplot
ins_bins_plot <- ggplot(data = aq_ins_clean_bins,
                        mapping = aes(x = bins, # x-axis: DO bins
                                      y = log_ave_lit, # y-axis: log_ave_lit
                                      color = taxon)) + # color by taxon

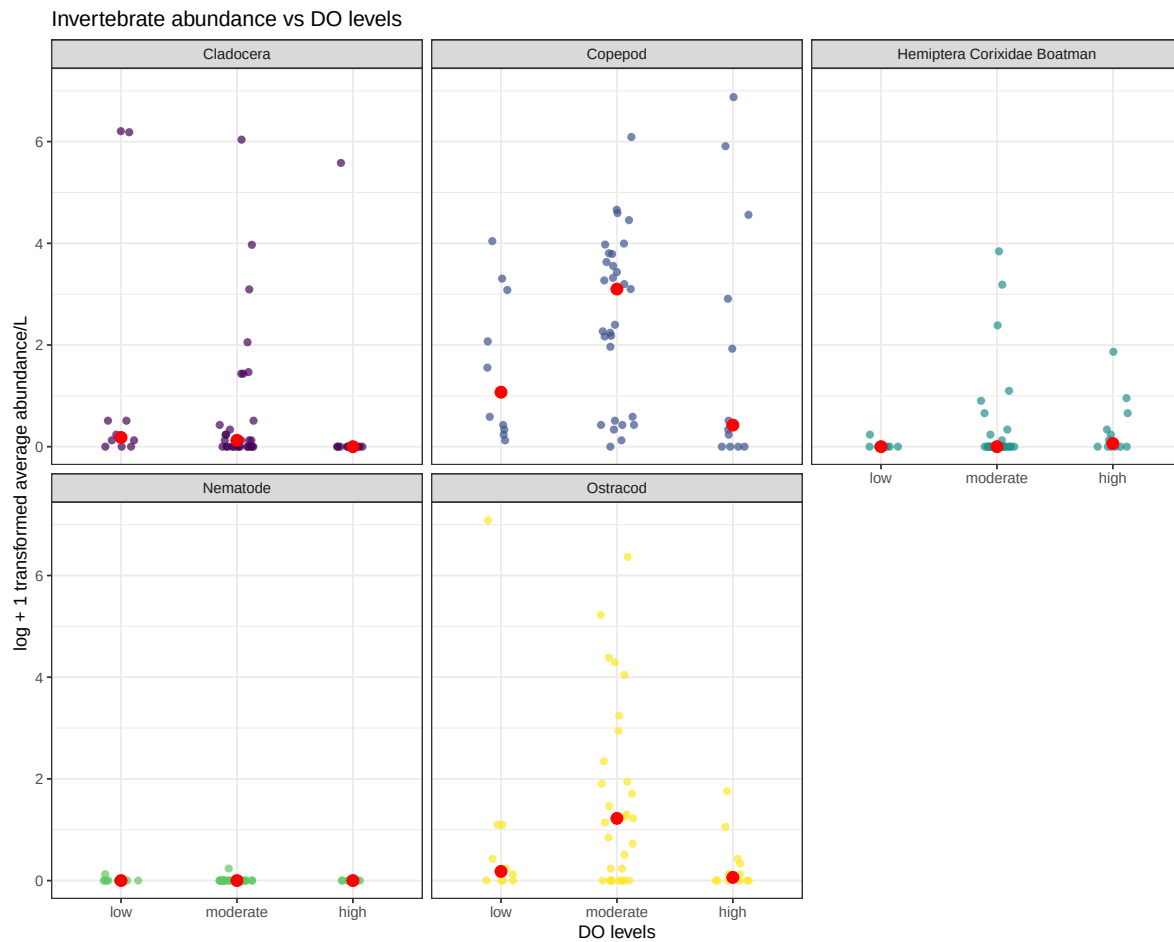
# first layer: points show individual abundance measurements
geom_jitter(width = 0.15,
            height = 0,
            shape = 16,
            size = 2,
            alpha = 0.7) +
# calculate median for each bin for each taxon
stat_summary(fun = median,
            geom = "point",
            color = "red",
            size = 3) +
# separate panels for each taxon
facet_wrap(~ taxon) +
# use non-default colors for points
scale_color_viridis_d() +
labs(x = "DO levels", # relabel x-axis
```

```

y = "log + 1 transformed average abundance/L", # relabel y-axis
# creates descriptive title
title = "Invertebrate abundance vs DO levels") +
# uses non-default theme
theme_bw() +
# remove redundant legend
theme(legend.position = "none")

# shows plot
ins_bins_plot

```



## **Plan for elective (if completing)**

- should include some description/image of drafts
- should include plan for making progress in the next two weeks